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Earth and Planetary Science Letters 172 (1999) 273–285

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Paleomagnetism of the Paleoproterozoic hematitic breccia and paleosol at Ville-Marie, Québec: further evidence for the low paleolatitude of Huronian glaciation

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Received 10 May 1999; revised version received 12 August 1999; accepted 12 August 1999

Abstract

The paleomagnetism of a saprolitic paleosol developed on Archean granite and of an overlying hematitic breccia in the basal Lorrain Formation of the Cobalt Group (Huronian Supergroup, ~2.4–2.3 Ga) near Ville-Marie, Québec, has been investigated to further constrain the paleolatitude of the Lorrain Formation and the subjacent glaciogenic Gowganda Formation. Only the breccia, whose hematite may be a product of weathering prior to or shortly after deposition, yielded useable results with stepwise thermal demagnetization. A stable component A (maximum unblocking temperature, T_{ub}, of ~675ºC) carried by hematite is directed shallowly to the east-northeast ($D = 59.5^{o}$, $I = 2.4^{o}$, $\alpha_{95} = 6.7^{o}$, $n = 47$, dip-corrected), and a less stable component B (maximum T_{ub} of ~580ºC) carried by magnetite is directed steeply down to the northeast ($D = 52.1^{o}$, $I = 70.3^{o}$, $\alpha_{95} = 2.8^{o}$, $n = 75$, dip-corrected). Component A is ascribed to a chemical remanent magnetization (CRM) acquired when the hematitic breccia was deposited or soon thereafter. Component B is interpreted as a thermochemical overprint related to the Penokean Orogeny at 1.9–1.8 Ga. Dip-corrected specimen mean directions give the pole positions at latitude = 57.1ºN, longitude = 338.3ºE (dp = 4.1º, dm = 4.8º) for component B, and latitude = 21.1ºS, longitude = 213.1ºE (dp = 3.3º, dm = 6.7º) for component A. The more easterly directions for components A and B relative to comparable early and overprint components previously determined for the Lorrain and Gowganda formations at Elliot Lake and Desbarats 260–360 km southwest of Ville-Marie imply relative rotation of ~30º or more about a vertical axis within the Southern Province since the Penokean Orogeny. The results suggest that the hematitic breccia and the Ville-Marie paleosol formed at a paleolatitude of $1.2^{o} \pm 3.4^{o}$, in accord with our earlier inference of equatorial paleolatitudes for the Lorrain and Gowganda formations and Huronian glaciation. Paleomagnetic data from several continents suggest that ferruginous weathering horizons formed over a wide range of paleolatitudes in the Proterozoic. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: Precambrian; glaciation; Paleosols; paleomagnetism; Huronian; Lorrain Formation

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PII: S0012-821X(99)00201-0

1. Introduction

The paleolatitudes of glaciation during the Neoproterozoic [1–6] and Paleoproterozoic [7,8] and the inferred glacial paleogeographies [9–13] are among the most contentious topics in contemporary Earth science. The paleomagnetism of the glaciogenic Gowganda Formation of the Paleoproterozoic Cobalt Group, Huronian Supergroup [14,15], in the Canadian Southern Province is critical to these debates, but several investigations have yielded different interpretations that imply conflicting paleogeographies. Early studies [16–18] suggested that the Gowganda Formation was deposited in high to moderate paleolatitudes, whereas our recent study [8] showed that more comprehensive data better accord with a near-equatorial origin for the Gowganda Formation.

In general, Huronian strata are not sufficiently deformed to provide unambiguous fold tests on all components, although components with moderate inclination and some with high inclination that are well defined have been shown to be post-folding in age [8,18,19]. The paleomagnetism of the arenaceous Lorrain Formation, which conformably overlies the Gowganda Formation, is central to the issue because Lorrain red beds near Desbarats contain magnetite and hematite that have retained six distinct components with low, moderate and high inclinations [8]. These remarkable red beds evidently have recorded several geological events since their deposition while retaining a stable remanence that is interpreted as being of early diagenetic origin and dating from the time of lithification. Further evidence for the relative timing of low- and high-inclination components displayed by the Lorrain Formation would be important in constraining the paleolatitude not only of the Lorrain Formation itself but also of the Gowganda Formation and Huronian glaciation.

Here we report the paleomagnetism of a hematitic breccia evidently of weathering and mass-flow origin in the basal Lorrain Formation and of an underlying saprolitic paleosol developed on Archean granite near Ville-Marie, southwestern Québec, to further constrain the paleolatitude of the Lorrain and Gowganda formations. The paleomagnetic study of Phanerozoic paleosols is well established [20,21], and we have demonstrated the suitability for paleomagnetic study also of Neoproterozoic hematitic paleosols [22] and a Proterozoic hematitic debris-flow breccia derived from a ferruginous regolith [23]. The present results support our earlier conclusion [8] of equatorial paleolatitudes for the Lorrain and Gowganda formations and for Huronian glaciation.

2. Geology

The age of the Huronian Supergroup in the Southern Province of Ontario and Québec is constrained by the age of 2473^{+16}_{-9} Ma for the Matachewan mafic dyke swarm that intrudes basement rocks of the Superior Province [24], the ages of 2491 ± 5 and 2480^{+10}_{-5} for intrusions believed to be co-magmatic with flood basalts of the basal Huronian succession [25] and 2450^{+25}_{-10} Ma for rhyolite near the top of the Huronian flood basalts [25], and the age of 2217–2210 Ma for the Nipissing Diabase that intrudes the Huronian Supergroup [26]. The Penokean Orogeny at 1900–1800 Ma produced open folds in Huronian strata in the northern part of the Southern Province [27,28], where the strata usually have low dips and the regional metamorphic grade does not exceed lower greenschist [29]. Although diabase intrusions have affected Huronian strata in many places, the effects of such intrusions are not evident in the Ville-Marie area of Québec.

The Ville-Marie paleosol [30] is developed on Archean granite beneath near-horizontal strata of the Lorrain Formation 4 km northwest of Ville-Marie, Québec (Fig. 1; paleosol at 47.35ºN, 280.51ºE) where the Gowganda Formation pinches out against Archean basement paleotopography. Rainbird et al. [30] divided the Ville-Marie paleosol into three zones (Fig. 2).

(1) At the base, a ‘protolith’ of unaltered or little-weathered, coarse-grained equigranular biotite granite of late Archean (Algoman) age. Two samples of the protolith contain 0.60% and 0.90% Fe_2O_3, 0.76% and 0.66% FeO, and 5.30% and 5.00% K_2O [30].

(2) A reddish to yellowish ‘transition zone’ 1–2 m thick in which plagioclase is almost completely sericitized, albite is partially sericitized, and biotite is altered to iron-rich chlorite. Microcline is unaltered. One sample contains 1.6% Fe_2O_3, 0.51% FeO, and 6.74% K_2O [30].

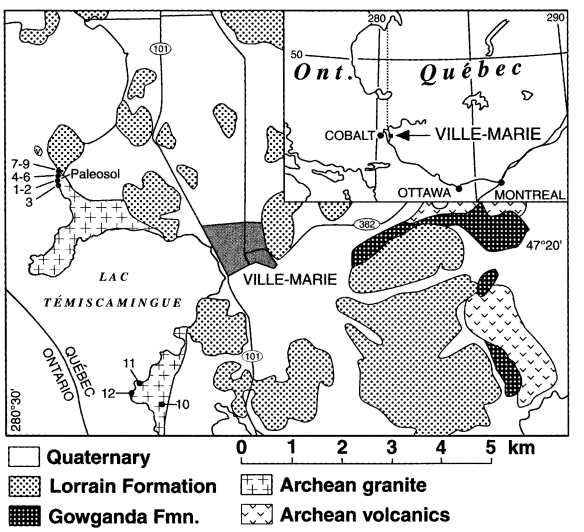


Fig. 1. Geological sketch map of the Ville-Marie area, Québec (adapted from [30]), showing the localities of paleomagnetic sites LP*01–12*.

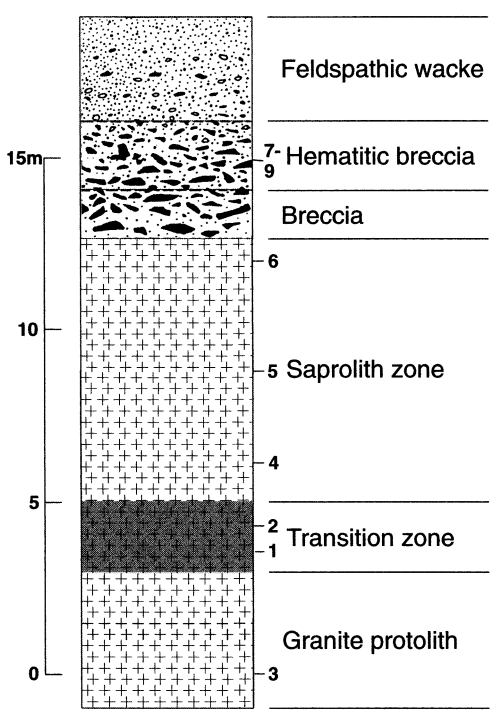


Fig. 2. Profile of the Ville-Marie paleosol and stratigraphy of the overlying basal Lorrain Formation (adapted from [30]), showing the approximate positions of paleomagnetic sites LP*01–09* (right side of column).

(3) At the top, a yellowish-green 'saprolith zone' ~7.5 m thick that preserves a granitic texture, although plagioclase and albite are replaced by sericite, microcline is partly sericitized, and quartz shows partial dissolution. Six samples taken across the saprolith zone contain 1.24–1.83% Fe_2O_3, 0.38–0.55% FeO, and 8.17–8.84% K_2O [30]. The texture, mineralogy and geochemistry of the saprolith zone are ascribable to primary weathering in part modified by diagenetic reactions.

An overlying, crudely bedded breccia at the base of the Lorrain Formation is 4 m thick. The lower 2 m comprises angular clasts 1–30 cm in length of locally derived dacite set in a matrix containing granitic detritus. Two samples from this lower breccia zone contain 3.79% and 4.94% Fe_2O_3, 0.33% and 0.78% FeO, and 8.70% and 7.40% K_2O [30]. The upper 2 m of the breccia is dusky red (5R 3/4) due to the presence of abundant ultrafine-grained hematite throughout the matrix, which supports angular clasts of completely sericitized dacite and scattered grains of weathered granite (all color designations given here are from the Geological Society of America Rock-Color Chart, 1963). Most clasts of altered dacite are greenish gray (5G 6/1) to bluish gray (5B 6/1), a few have light brown (5YR 7/4) rims and mottles, and rare clasts have thin (1–2 mm), discontinuous hematitic cutans or skins that evidently were eroded partially before burial. Rainbird et al. [30] reported that two samples from the hematitic breccia contain 5.29% and 6.17% Fe_2O_3, 0.84% and 4.86% FeO, and 8.96% and 5.69% K_2O. A sample we collected of hematitic matrix with scattered sand-grade granitic detritus contains 18.47% Fe_2O_3 and 1.41% FeO, as well as 49.58% SiO_2, 17.50% Al_2O_3, 6.47% K_2O, 1.99% MgO and 0.70% TiO_2; other major elements are <0.05%.

Rainbird et al. [30] interpreted the breccia unit as "a locally derived mass flow deposit which infilled a topographic depression where the saprolith was preserved." They further suggested that the hematitic breccia zone "may represent the upper portion of a second weathering profile developed on the basal Lorrain Formation breccia." However, the presence of differently altered dacite clasts and of partially eroded cutans suggests to us that the material constituting the hematitic breccia was weathered before slumping to its site of deposition. Hence the origin

of the hematitic breccia may be similar to that of the Gangau diamictite at the base of the Proterozoic Vindhyan Supergroup in central India, which contains up to 20% Fe_2O_3 in its hematitic matrix and is interpreted as a mass-flow deposit derived from a ferruginous regolith [23]. The hematitic breccia grades upward, with no sign of any hiatus, through cross-bedded pink feldspathic wacke to subarkose of the Lorrain Formation and hence, by either interpretation of the breccia, its ferruginous matrix formed near the time of deposition. The mineralogy of the hematitic breccia accords with that of 'Type A red beds' [31], which have fairly mature mineralogy at the time of deposition and whose remanence is an early chemical remanent magnetization (CRM). The presence of these red beds immediately above the Ville-Marie paleosol suggests that the atmosphere contained an unquantified but substantial amount of oxygen by ~2300 Ma [32].

3. Paleomagnetism

3.1. Previous studies of the Huronian Supergroup

Previous paleomagnetic studies of the Huronian Supergroup are reviewed by Williams and Schmidt [8]. The interpretation of results is made difficult because of widespread overprinting caused by diabase intrusions and regional deformation and the paucity of red beds that may carry stable remanence. For many years the Morris A direction ($D = 29^o$, $I = 52^o$, dip-corrected [18]) for gray argillites from the glaciogenic Coleman Member of the Gowganda Formation and the Roy and Lapointe B direction ($D = 340^o$, $I = 56^o$, in situ [17]) for grayish red argillite of the deltaic Firstbrook Member of the Gowganda Formation were widely regarded as having been acquired soon after deposition. However, Buchan et al. [33] found that gray units of the Gowganda Formation are unsuitable for determining primary directions, thus casting doubt on earlier results. Subsequently, Williams and Schmidt [8] carried out a comprehensive paleomagnetic study of the Gowganda Formation at Elliot Lake 260 km southwest of Ville-Marie and of the Lorrain Formation at Desbarats 360 km southwest of Ville-Marie. They found stable components with low inclinations carried by hematite for sandstones of the Coleman Member at Elliot Lake (Coleman sandstone A: $D = 216.3^o$, $I = 5.5^o$, dip-corrected) and for a red bed unit of the Lorrain Formation at Desbarats (Lorrain A: $D = 5.4^o$, $I = 5.5^o$, dip-corrected) that they ascribed to early CRM acquired fairly soon after deposition. Williams and Schmidt [8] re-interpreted the Morris A component [18] as an overprint related to the Penokean Orogeny. The present study of the basal Lorrain hematitic breccia and the Ville-Marie paleosol has been conducted to test the inference that the components with shallow directions were acquired early.

3.2. Sampling and laboratory techniques

Site localities are shown in Fig. 1. Samples of the granite protolith, the Ville-Marie paleosol, and hematitic breccia of the Lorrain Formation were collected using a portable drill, whereas fresh granite from farther south was block sampled. Orientation was carried out using a tool that accommodates both a sun compass and a magnetic compass. The mean of the local declination thus measured agreed within error with the published value of 12°W.

The stratigraphic positions of sites LP01–09 are shown in Fig. 2. The Ville-Marie paleosol and Lorrain Formation were sampled along a traverse at the eastern shore of Lac Témiscamingue (Lake Timiskaming) that included the granite protolith (site LP03), the transition zone (LP01, LP02), the saprolith zone (LP04–06), and the hematitic breccia (LP07–09). Site LP07 is located directly above the exposed granite–breccia unconformity, and sites LP08 and LP09 are located ~20 m and ~40 m farther north, respectively. The structural attitude of the hematitic breccia was taken as 6° dip toward 353°, based on bedding within and at the top of the breccia. In addition, fresh Algoman granite was sampled at three sites 4 km south of the paleosol exposure: in a road cut (LP10), in a quarry (LP11) and from a cliff face (LP12).

Routine paleomagnetic laboratory methods [34, 35] were employed. Remanent magnetizations were measured initially with a CTF three-axis RF SQUID magnetometer and subsequently with a 2G-775R three-axis DC SQUID magnetometer interfaced to a computer. The 2G system incorporates an automatic, alternating field (AF) demagnetizer. The CSIRO au-

tomated three-stage carousel furnace was used for thermal demagnetization. The furnace is housed inside a 4 m ten-coil Helmholtz set with automatic feedback maintaining zero-field (<5 nT). All specimens were subjected to stepwise thermal demagnetization, and some specimens also were subjected to stepwise AF demagnetization.

Susceptibility versus temperature measurements were conducted using a high-sensitivity variable-field translation balance (VFTB). Small (5 mm) cylindrical slugs were cored from off-cuts of standard-sized paleomagnetic specimens for these measurements.

Components of magnetization were isolated using an interactive version of LINEFIND [36]. Because there is evidence that some specimens carry remanence of both normal and reverse polarities, we have assigned unit weight to specimens in the statistical analysis.

3.3. Results

The specimens of fresh granite (LP10–12) had a mean intensity of natural remanent magnetization (NRM) of 50 ± 35 mA m^{-1}. The distribution of NRM directions reflects the axial directions of cores drilled in the laboratory, suggesting the predominance of unstable magnetic carriers in the granite. Magnetic susceptibilities were around 6.0×10^{-3} SI although extremes reached as low as 0.3×10^{-3} SI and as high as 15×10^{-3} SI.

Compared to the fresh granite, the NRM intensities and the susceptibilities of protolith and transition zone specimens (LP01–03) were low, being 5.8 ± 3.0 mA m^{-1} and $0.5 \times 10^{-3} \pm 0.16 \times 10^{-3}$ SI, respectively. NRM intensities of the saprolith specimens (LP04–06) were even weaker, with a mean of 0.9 ± 0.3 mA m^{-1}. Saprolith NRM directions are scattered around the present field direction (or dipole direction) at the locality. The magnetic susceptibilities of the saprolith specimens ($0.1 \times 10^{-3} \pm 0.02 \times 10^{-3}$ SI) also were lower than their protolith–transition zone equivalents, perhaps indicating the preferential leaching of magnetite. On stepwise thermal demagnetization, specimens from paleosol sites (protolith–transition zone–saprolith zone, LP01–06) showed no systematic trends and at high temperatures the remanence directions of individual specimens behaved erratically. Apparently, these sites have not retained a stable ancient remanence.

Specimens of hematitic breccia (LP07–09) had a NRM mean intensity of 10 ± 5 mA m^{-1} and a mean susceptibility of 1.0×10^{-3} SI, with extremes as high as 2.5×10^{-3} SI and as low as 0.4×10^{-3} SI. The breccia NRMs are fairly well grouped in the northeast quadrant, moderately to steeply downward-directed. The distribution of NRM directions from site LP08 shows streaking towards a shallow direction to the northeast. By contrast, seventeen specimens taken from clasts of altered dacite in the hematitic breccia had a mean NRM intensity of only 1.7 ± 1.1 mA m^{-1} and a mean susceptibility of $0.06 \times 10^{-3} \pm 0.04 \times 10^{-3}$ SI.

The variation of susceptibility (in a moderate field of ~48 kA m^{-1} or 600 Oe) with temperature for a specimen of the hematitic breccia (LP09j) is plotted as induced magnetization versus temperature in Fig. 3a. On heating, two Curie temperatures are evident, one at ~580ºC and another at ~675ºC. These two temperatures would appear to indicate the presence of magnetite and hematite, respectively. Although the magnetite contribution to the induced magnetization is about twice that of the hematite, volumetrically the specimen is dominated by hematite, with magnetite forming only a trace amount. The curve is irreversible and shows a three-fold increase in magnetization on cooling. The magnetization (in a field of 614 kA m^{-1} or 7720 Oe) versus temperature plot (Fig. 3b) for a clast of altered dacite shows that the magnetic properties are dominated by paramagnetic material plus a very small fraction of material with an ill-defined Curie temperature between 550ºC and 600ºC, probably magnetite. The dotted curves shown in Fig. 3b represent the ferromagnetic (sensu lato) component of the clasts and were derived by fitting and subtracting the paramagnetic contribution using a function that varies inversely with (absolute) temperature. If the ferromagnetic material is magnetite it would be saturated in the applied field, yet the magnetization is only 10 mA m^{-1} (10^{-5} emu cm^{-3}). Because the saturation magnetization of magnetite is 490 kA m^{-1} (490 emu cm^{-3}) it follows that the concentration of magnetite in the clasts is 0.02 ppm. This finding further suggests that the clasts of altered dacite are unsuitable for paleomagnetic analysis.

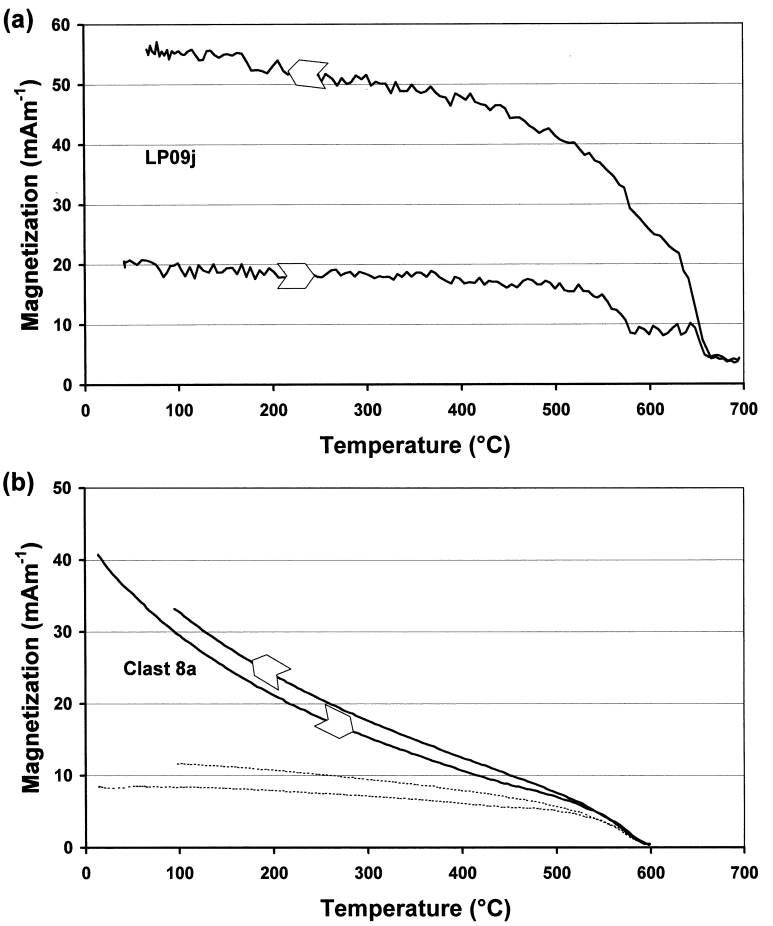


Fig. 3. Variation of magnetization with temperature using a variable field translation balance (VFTB) for: (a) a specimen of hematitic breccia (*LP09j*) in a field of 47.7 kA m^{-1} or 600 Oe; and (b) a specimen from a clast of altered dacite (LP08*a*) in a field of 614 kA m^{-1} or 7720 Oe. The arrows show the heating and cooling curves. The dotted lines in (b) represent the ferromagnetic (magnetite?) component of the clast, after fitting and subtracting the paramagnetic contribution (see text).

Only the hematitic breccia yielded useable results on demagnetization. Thermal demagnetization was found to be the most effective method to resolve the magnetic components present in this unit; AF demagnetization had little effect on the remanence of hematitic breccia specimens. The stability to AF demagnetization and the high Curie temperature, determined on the VFTB and inferred from the maximum T_{ub} (unblocking temperature) on thermal demagnetization, implies that hematite is the remanence carrier.

Most directions in the hematitic breccia became shallower on stepwise thermal demagnetization until ~630ºC, where the direction stabilized and demagnetized toward the origin (Fig. 4). The orthogonal projections show that an additional component is present, especially in specimens from site LP08. About 80% of specimens from this site revealed a component directed westerly and downward that was removed in varying amount between 600ºC and 630ºC. This component may be present in specimens from sites LP07 and LP09 but was not resolved by thermal demagnetization because of overlapping T_{ub} spectra. Although it is not possible to delineate this component accurately, it is interpreted as a small parasitic reverse component carried by hematite. Reversals are common in the products of weathering such as laterites and paleosols [20] and are evidence

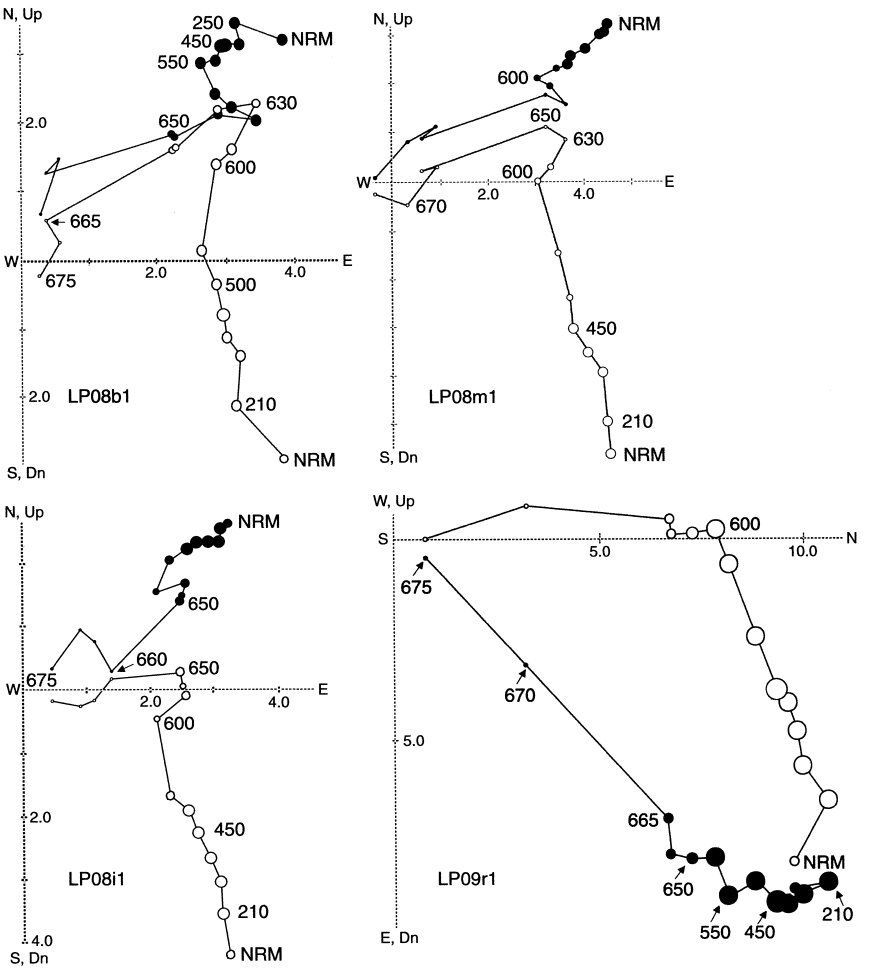


Fig. 4. Orthogonal projections showing examples of behavior on stepwise thermal demagnetization, from NRM to 675ºC, of specimens of hematitic breccia from the basal Lorrain Formation near Ville-Marie. Dots represent magnetization vectors projected onto the horizontal plane and circles represent projections onto the vertical plane. *Dn* = down. Circle size indicates the circular standard deviation of each measurement. Units are mA m^{-1} (= 1 μG = 10^{-6} emu cm^{-3}). Specimen numbers refer to site/sample/specimen.

that the magnetization is a CRM and that secular variation probably has been averaged.

The least stable component B (Fig. 5a) present in the hematitic breccia is directed steeply down to the northeast ($D = 52.1^{o}$, $I = 70.3^{o}$, $\alpha_{95} = 2.8^{o}$, $n = 75$, dip-corrected), whereas the most stable component A (Fig. 5b) is directed shallowly to the east-northeast ($D = 59.5^{o}$, $I = 2.4^{o}$, $\alpha_{95} = 6.7^{o}$, $n = 47$, dip-corrected). Component A evidently is carried by hematite, as identified on the VFTB, and is best ascribed to CRM acquired when the hematitic breccia was deposited or soon thereafter. The ease with which component B is removed on thermal demagnetization suggests that it is carried by magnetite, which again is consistent with the VFTB measurements, and B probably is a thermochemical overprint related to an orogenic event. Hence the hematitic breccia is characterized by two predominant components, one probably early and one much later, and a small parasitic component of unidentifiable origin, but apparently carried by hematite.

In the absence of a fold test we sought to execute a conglomerate test. The magnetic properties of the clasts of altered dacite, which are drab in color, differ

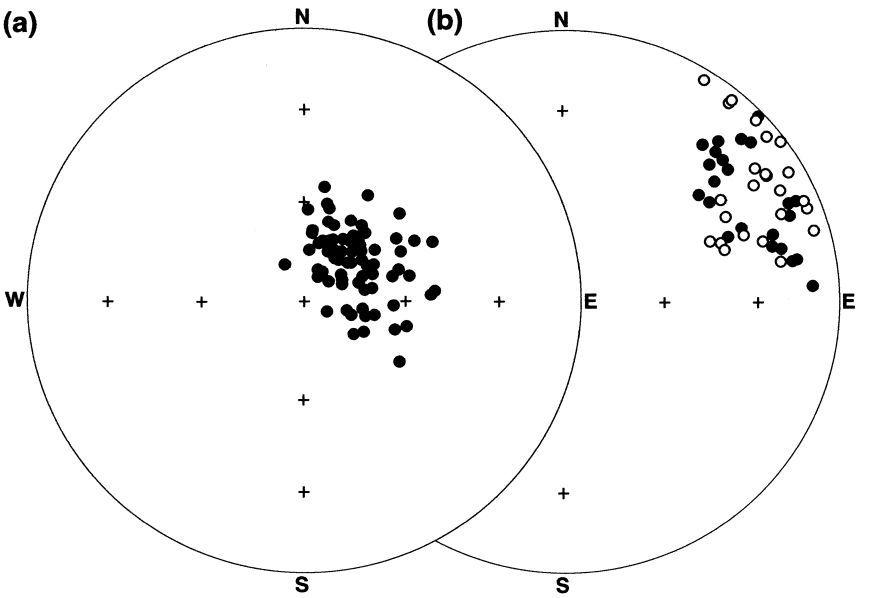


Fig. 5. Equal-area stereographic projections of dip-corrected directions for: (a) the lower unblocking temperature component B (maximum T_{ub} of ~580ºC); and (b) the high unblocking temperature component A (maximum T_{ub} of ~675ºC) isolated for specimens of the hematitic breccia from the basal Lorrain Formation. Dots indicate down directions and circles up directions.

greatly from the breccia matrix, where virtually all of the hematite resides: the mean NRM intensity and the mean susceptibility of these clasts are an order of magnitude smaller than those of the hematitic breccia, and the magnetization versus temperature plot (Fig. 3b) indicates a lack of suitable paleomagnetic carriers. Hence the clasts themselves could not provide a conglomerate test. The rare cutans on clasts of altered dacite also were considered, but because the cutans and the ferruginous matrix evidently formed closely in time it is possible that they both acquired the same CRM subsequent to deposition, in which case a conglomerate test would be inconclusive. On the other hand, if the magnetization of the cutans could be shown to pre-date deposition, it would argue against a blanket remagnetization at some time after deposition. However, it was not practicable to isolate the magnetic properties of the few, thin cutans in the oriented cores that we collected.

Table 1 lists the in situ and dip-corrected site and specimen mean directions for components A and B in the hematitic breccia. Using dip-corrected specimen mean directions, the corresponding pole positions are latitude = 57.1ºN, longitude = 338.3ºE (dp = 4.1º, dm = 4.8º) for the lower unblocking temperature component B (maximum T_{ub} of ~580ºC), and latitude = 21.1ºS, longitude = 213.1ºE (dp = 3.3º, dm = 6.7º) for the high-temperature component A (maximum T_{ub} of ~675ºC).

The circular symmetry of component A was checked using the modified Kuiper statistic ([37], p. 114). Testing the distribution with respect to the Fisher mean direction given in Table 1, the following values were obtained: $D_n^+ = 0.087$, $D_n^- = 0.082$, leading to a modified Kuiper statistic $V_n^* = 1.191$. This figure is less than the 85% confidence point of 1.537 listed by Fisher et al. ([37], table A6) and indicates that the null hypothesis of circular symmetry cannot be rejected at the usual 95% level.

4. Discussion

Table 2 gives directions of hematitic breccia components A and B (this study) and of comparable components for the Lorrain Formation at Desbarats (Lorrain A and B [8]) and the Coleman Member of the Gowganda Formation at Elliot Lake (Coleman sandstone A [8] and Morris A [18]). Paleomagnetic poles for these six components are plotted on a

Table 1
Paleomagnetic results for the hematitic breccia of the basal Lorrain Formation, Ville-Marie, Québec (47.35ºN, 280.51ºE)

Site	n/N	D_h (º)	I_h (º)	D_b (º)	I_b (º)	k	α_{95} (º)
Component B (maximum $T_{ub} \approx 580$ºC)							
07	22/13	83.7	73.4	64.2	72.5	40.8	4.9
08	25/20	57.2	73.2	42.1	69.9	34.0	5.0
09	28/28	67.5	70.8	52.6	68.5	32.6	4.8
Mean	75/61	68.7	72.7	52.1	70.3	34.1	2.8
Pole B (dip-corrected): latitude = 57.1ºN, longitude = 338.3ºE (dp = 4.1º, dm = 4.8º)							
Component A (maximum $T_{ub} \approx 675$ºC)							
07	5/13	67.1	15.0	65.6	13.3	10.3	25.0
08	20/20	57.7	−6.8	58.5	−9.4	11.6	10.0
09	22/28	60.2	12.8	59.0	10.4	13.1	8.9
Mean	47/61	59.8	4.7	59.5	2.4	10.7	6.7
Pole A (dip-corrected): latitude = 21.1ºS, longitude = 213.1ºE (dp = 3.3º, dm = 6.7º)							

n = number of specimens; N = number of cores collected; D_h, I_h = mean declination and mean inclination with respect to present horizontal; D_b, I_b = mean declination and mean inclination with respect to bedding; k = precision parameter; α_{95} = semi-angle of cone of 95% confidence about mean direction; dp, dm = semi-minor and semi-major axes of polar error ellipse.

Table 2
Relative rotation of declinations (dip-corrected) of components for the Lorrain Formation at Desbarats, the Coleman Member of the Gowganda Formation at Elliot Lake, and the Lorrain hematitic breccia at Ville-Marie

Component	D_b (º)	I_b (º)	α_{95} (º)	ΔD (º)	Relative rotation (º)
Overprint					
Lorrain B, Desbarats [8]	18.6	59.6	6.1	12.1	5.4 ± 23.0
Morris A, Elliot Lake [18]	24	54	15	26.1	28.1 ± 21.9
Hematitic breccia B, Ville-Marie (this study)	52.1	70.3	2.8	8.3	
Early					
Lorrain A, Desbarats [8]	5.4	5.5	5.9	5.9	30.9 ± 7.1
Coleman sandstone A, Elliot Lake [8]	216.3	5.5	6.7	6.7	23.2 ± 7.5
Hematitic breccia A, Ville-Marie (this study)	59.5	2.4	6.7	6.7	

D_b, I_b = mean declination and mean inclination with respect to bedding; α_{95} = semi-angle of cone of 95% confidence about mean direction, and ΔD the corresponding declination error. The rotation errors were calculated following Butler ([35], eq. A.44 and eq. A.66).

global projection in Fig. 6 together with ∼2.5–2.0 Ga primary paleopoles for Paleoproterozoic dyke swarms from the Canadian Superior and Southern provinces [38].

The interpretation that the stable, high-temperature, shallowly directed hematitic breccia A was acquired early accords with our previous findings that stable, high-temperature, shallowly directed Lorrain A and Coleman sandstone A evidently were acquired near the time of deposition [8]. The discrete poles for these three early components for the Cobalt Group form an arcuate trend near the primary poles for the older (∼2.50–2.45 Ga) Ptarmigan and Matachewan dyke swarms (Fig. 6). The proximity of poles for the Cobalt Group and these basement dykes may indicate a relatively short duration of Huronian deposition and/or a quasi-static interval in the Canadian Proterozoic apparent polar wander path.

Furthermore, hematitic breccia B is comparable to Lorrain B: these components are directed steeply down to the northeast and north-northeast, respectively, and are interpreted as overprints related to

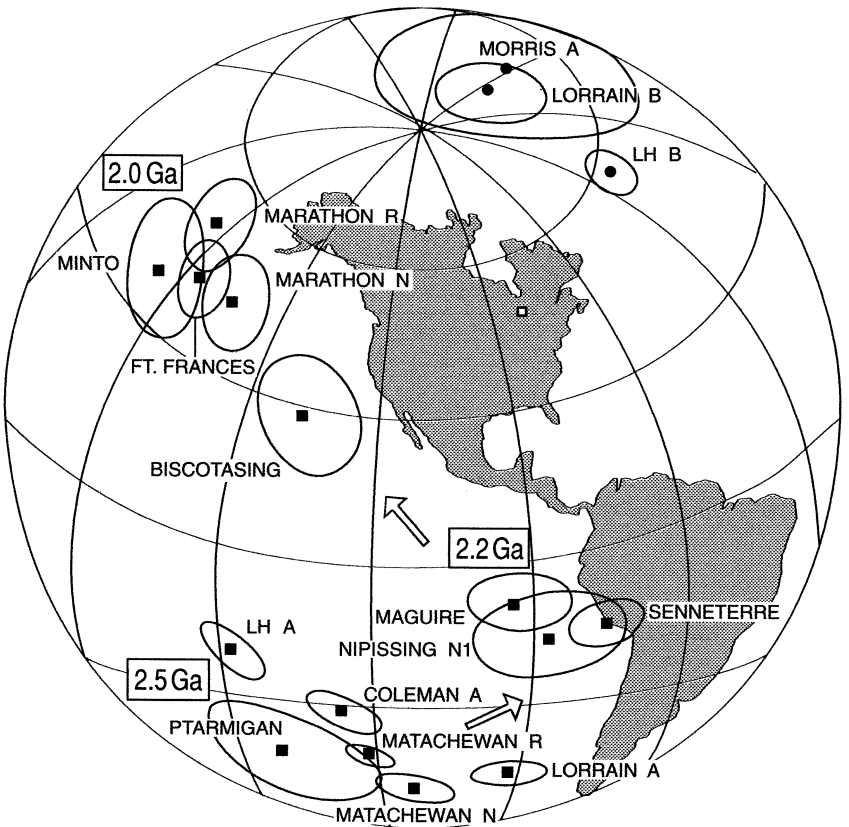


Fig. 6. Primary paleomagnetic poles for Lorrain hematitic breccia A (*LH A*, this study), Lorrain A and Coleman sandstone A [8], and ~2.5–2.0 Ga primary paleopoles for Paleoproterozoic dyke swarms from the Canadian Superior and Southern provinces (Ptarmigan, Matachewan normal and reverse, Nipissing N1, Senneterre, Maguire, Biscotasing, Marathon normal and reverse, Fort Frances, and Minto [38]). Also shown are overprint poles for Lorrain hematitic breccia (*LH B*, this study), Lorrain B [8] and Morris A [18]. Polarities are not implied. The sampling area for the present study is shown as a rectangle.

orogenic events ([8] and this study). In addition, Morris A, which is interpreted as an overprint related to the Penokean Orogeny at 1.9–1.8 Ga [8], is directed steeply down to the north-northeast. Because Morris A pre-dates folding [18], hematitic breccia B and Lorrain B also may have been acquired just prior to Penokean folding. Hence dip-corrected directions are given in Table 2 and have been used to calculate the respective pole positions shown in Fig. 6. A reverse component Lorrain E, whose in situ direction is closely anti-parallel to dip-corrected Lorrain B, may have been acquired just after folding [8].

These sets of early and overprint components show consistent changes in direction across the Southern Province. The declinations for the three early components for the Cobalt Group are directed progressively more to the east, proceeding eastward over a distance of 360 km from Desbarats to Elliot Lake to Ville-Marie, with differences in declination of $30.9^{o} \pm 7.1^{o}$ and $23.2^{o} \pm 7.5^{o}$ between respective areas (Table 2). The spread of the poles (Fig. 6) is reminiscent of that displayed by Nipissing N1 poles, which have been interpreted in terms of relative rotations about a vertical axis within the Southern Province, including relative rotation of up to 40° between the Blind River area in the west and the Cobalt area in the east [39]. Furthermore, McGlynn and Irving [40] suggested from paleomagnetic evidence that a local rotation of 60° occurred within the Paleoproterozoic Athapuscow Aulacogen in northwestern Canada during the Proterozoic. More recently, Bates and Halls [41] provided evidence that

broad-scale vertical-axis relative rotations of up to 44°, as recorded by the spread of Matachewan dyke declinations and accompanying changes in trend, affected the Superior Province during the Proterozoic. Our data provide further evidence for vertical-axis rotation within the Southern Province, and suggest that the hematitic breccia A pole from the block in the eastern part of that province may have undergone relative clockwise rotation. Restoring this block by an anticlockwise rotation of ~30° or more brings the three Cobalt Group poles into closer proximity.

Declinations for the three overprint components also are directed progressively more to the east proceeding eastward from Desbarats to Elliot Lake to Ville-Marie, with differences in declination of 5.4° ± 23.0° and 28.1° ± 21.9° between respective areas (Table 2). Although these rotations are less definitive than those for the low-inclination early directions, it is significant that a ~30° restoration as suggested above for hematitic breccia A brings the hematitic breccia B pole into close proximity to the poles for Lorrain B and Morris A (Fig. 6). The suggested rotation would have occurred since the Penokean Orogeny.

As we noted previously [8], the difference in declinations for Lorrain A and Coleman sandstone A may partly reflect movement on the Murray Fault Zone, an important and long-lived Proterozoic fault system that runs approximately east–west between Desbarats and Elliot Lake within the Southern Province west of Sudbury [42,43]. Similarly, the difference in declinations for Coleman sandstone A and hematitic breccia A may partly reflect movement on the system of major faults forming the northwest-trending Lake Timiskaming Rift Valley [44] that is centered on Lake Timiskaming immediately west of Ville-Marie in the eastern part of the Southern Province. Our sampling site near Ville-Marie occurs within the rift ~3–35 km northeast of the main faults, which extend en echelon for more than 300 km, have vertical displacements of up to 3 km, and are thought to have controlled major Nipissing Diabase intrusions and mineralization in the Cobalt–Gowganda area ~20–100 km northwest of Ville-Marie [44]. Hence major fault systems with histories dating from the Paleoproterozoic separate the three areas of the Southern Province for which we have identified early paleomagnetic components in Huronian strata (Table 2). How the rotation identified here might fit in with the scheme of Bates and Halls [41] for the Matachewan dyke swarm in the Superior Province is, however, outside the scope of this contribution. Our data imply that local vertical-axis rotation causes some disruption of the Canadian Proterozoic apparent polar wander path, and we would agree that caution should be exercised when using paleomagnetic data for Archean and Paleoproterozoic terranes in paleotectonic reconstructions.

The present study suggests that the hematitic breccia of the basal Lorrain Formation formed at a paleolatitude of 1.2° ± 3.4°. Hence it provides support for our earlier preferred interpretation [8] that the Lorrain Formation and the Gowganda Formation, which is the most important glaciogenic succession of the Huronian Supergroup, accumulated within a few degrees of the paleo-equator. Neoproterozoic glaciogenic successions in Australia [2,3,5] and Canada [6] and evidently Paleoproterozoic tillite in South Africa [7] also formed near the paleo-equator.

Our results further imply that the weathering episodes which produced the Lorrain hematitic breccia and the underlying Ville-Marie saprolitic paleosol also occurred near the paleo-equator. Paleomagnetic data are available for few other Precambrian paleosols. Early Neoproterozoic saprolitic–hematitic paleosols in northwest Scotland developed on Lewisian Gneiss at a paleolatitude of $37.8^{\circ}{}^{+5.7^{\circ}}_{-5.0^{\circ}}$ [22]. The Proterozoic Gangau diamictite in India, which has sedimentological similarity to the Lorrain hematitic breccia and was derived from a ferruginous regolith, formed at a paleolatitude of $44.7^{\circ}{}^{+18.6^{\circ}}_{-13.5^{\circ}}$ [23]. Evidently ferruginous weathering horizons formed over a wide range of paleolatitudes during the Proterozoic.

5. Summary and conclusions

(1) A hematitic breccia in the basal Lorrain Formation of the Huronian Supergroup near Ville-Marie, Québec, yielded a stable component A (maximum T_{ub} of ~675°C) carried by hematite that is directed shallowly to the east-northeast ($D = 59.5°$, $I = 2.4°$, dip-corrected), and a less stable component B (maximum T_{ub} of ~580°C) that is directed

steeply down to the northeast ($D = 52.1^{o}$, $I = 70.3^{o}$, dip-corrected).

(2) Component A is ascribed to early CRM acquired near the time of deposition, and component B is interpreted as a thermochemical overprint related to the Penokean Orogeny at 1.9–1.8 Ga.

(3) The more easterly directions for components A and B relative to comparable early and overprint components for the Lorrain and Gowganda formations at Elliot Lake and Desbarats 260–360 km to the southwest imply clockwise rotation of $\sim 30^{o}$ or more of the Ville-Marie area relative to the other areas since the Penokean Orogeny.

(4) The present study suggests that the Lorrain hematitic breccia formed at a paleolatitude of $1.2^{o} \pm 3.4^{o}$. Hence the results are consistent with our earlier conclusion [8] that the Lorrain Formation and the conformably underlying Gowganda Formation, which is the most widespread glaciogenic succession of the Huronian Supergroup, accumulated near the paleo-equator.

(5) Our results affirm the suitability for paleomagnetic analysis of Precambrian hematitic deposits related to or derived from ferruginous weathering profiles, and imply that the weathering episodes which produced the Lorrain hematitic breccia and the Ville-Marie saprolitic paleosol also occurred near the paleo-equator. Together with other paleomagnetic data, the results suggest that Proterozoic ferruginous weathering horizons formed over a wide range of paleolatitudes.

Acknowledgements

We thank D. Clark for discussions, M. Huddleston for technical assistance, and H. Halls and D. Evans for reviewing the manuscript. Williams is supported by an Australian Research Council Senior Research Fellowship. ***[RV]***

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